INTERNATIONAL INSTITUTE OF INFORMATION TECHNOLOGY, HYDERABAD

RAG over the EU AI Act

Baseline Retrieval vs. Cross-Encoder Re-Ranking

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RAG Assignment

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Code: `https://github.com/vishakkashyapk30/lma-rag-assignment`

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Setup

I used the EU AI Act as my document. It is a 144 page PDF from the Official Journal of the EU. I chose it because it has many rules with exceptions. It also has rules that change with the role, like provider, deployer and importer. And it starts with 180 recitals. These are long explanations of the law. They are not the law itself, but they talk about the same things as the articles. So they can easily confuse a search system.

I cleaned the PDF text and cut it into 747 chunks. A chunk is one recital, or a few paragraphs of an article, or a point of an annex. No chunk is longer than 230 tokens, because the bi-encoder only reads 256. Every chunk starts with a short header, like `[Article 99 – Penalties, para 5–6]`.

The bi-encoder is `all-MiniLM-L6-v2` with a FAISS index. The cross-encoder is `ms-marco-MiniLM-L-6-v2`. The LLM is `qwen2.5:7b`, run with Ollama at temperature 0. I compare three setups. The *baseline* uses only the bi-encoder and sends its top 3 chunks to the LLM. *Re-ranking* takes the same top 10 chunks, orders them with the cross-encoder, and sends the new top 3. The *final pipeline* is an improved version that I describe in Section 5. All three use the same prompt. It says to answer only from the three chunks, to give the direct answer first, and to say so if the answer is not there. My first prompt only said to cite the headers. The model then often replied with just a header and nothing else. So I changed the prompt for all pipelines and made every answer again.

The code and results are in my GitHub repository: `https://github.com/vishakkashyapk30/lma-rag-assignment`.

I wrote 49 questions the way a normal person would ask them. For each one I marked the exact clause that holds the answer. Then I measured three things. *hit@3* asks if a right clause is in the top 3. *all@3* asks if every needed clause is in the top 3. *MRR@10* rewards a high rank. I also checked each answer for key facts, like "10 days". This check is rough, so I read the answers too. All 49 questions are on the last page.

1 Architecture Comparison

A bi-encoder turns the question into one vector. It turns each chunk into another vector. It does this separately. Then it compares the two with cosine similarity. The chunk vectors can be made once and saved, so a search is fast. But the question and the chunk never meet. One word like "not" or "importer" gets averaged with about 200 other words. It hardly changes the vector.

A cross-encoder reads the question and the chunk together as one input. Self-attention works across both of them, so every word of the question can look at every word of the chunk. It gives one score for the pair. This helps with small words and conditions. The cost is that nothing can be saved in advance. Every pair needs a new pass through the model.

Figure 1 shows the two pipelines side by side. Figure 2 shows how the two models see the same 490 candidates.

	Baseline	Re-ranking	Final
Setup			
Bi-encoder (first stage)	MiniLM-L6	MiniLM-L6	MiniLM-L6 + BM25
Cross-encoder (re-ranker)	none	MiniLM-L6	MiniLM-L12
Chunks re-ranked	none	10	30
Text searched	all	all	no recitals
Finding the right clause			
hit@3	0.633	0.653	**0.837**
all@3	0.612	0.633	**0.796**
MRR@10	0.474	0.502	**0.818**
MRR@10, Q01 to Q24	0.477	0.506	**0.812**
MRR@10, Q25 to Q49	0.471	0.498	**0.825**
Right clause among candidates	0.776	0.776	**0.980**
Answers			
Key facts found	0.673	0.769	**0.810**
Key facts, Q01 to Q24	0.604	0.715	**0.778**
Key facts, Q25 to Q49	0.740	0.820	**0.840**
Answers that gave up	3	3	3
Speed (median, ms)			
Retrieval	4.5	21.5	87.4
LLM answer	5,416	5,198	5,633

Table 1. All three setups on all 49 questions. Baseline is the bi-encoder alone. Re-ranking adds the cross-encoder on the top 10. Final is the improved pipeline from Section 5. "Right clause among candidates" is the share of questions where the right clause was in the top 10 (baseline, re-ranking) or in the 30 candidates (final). The best value in each row is in bold.

Question	Type	Clause	Rank
Q16 importer checks	role limit	Art 23(1)	#4 → #1
Q02 scraping photos	negation	Art 5(1)(e)	#4 → #2
Q27 sort by race	negation	Art 5(1)(g)	#5 → #2
Q47 fraud exception	exception	Annex III 5(b)	#5 → #2

Table 2. Clauses that moved from rank 4 or 5 into the top 3.

They agree on the big picture, with a Spearman correlation of 0.72. But inside the top 10 they often disagree. About one third of the bi-encoder's top 3 chunks were pushed out of the top 3 by the cross-encoder.

2 Qualitative & Rank Shift Analysis

Table 1 has the main numbers. Re-ranking is only a little better than the baseline on retrieval. The answers improved by more. The re-ranked answer had more key facts than the baseline answer for 7 questions, fewer for 1, and the same for the other 41. Figure 3 shows what happened to the right clause for every question. (The final pipeline is explained in Section 5.)

Where re-ranking helped. Four questions are clear wins (Table 2). In each one the right clause was at rank 4 or 5, and the cross-encoder moved it into the top 3.

Q16 asks what an importer must check. All the paragraphs of Article 23 use the same words, so the bi-encoder put paragraph 2 first. The baseline answer then said the importer must make sure the system "is not in conformity with the Regulation". That is the opposite of the rule. Af-

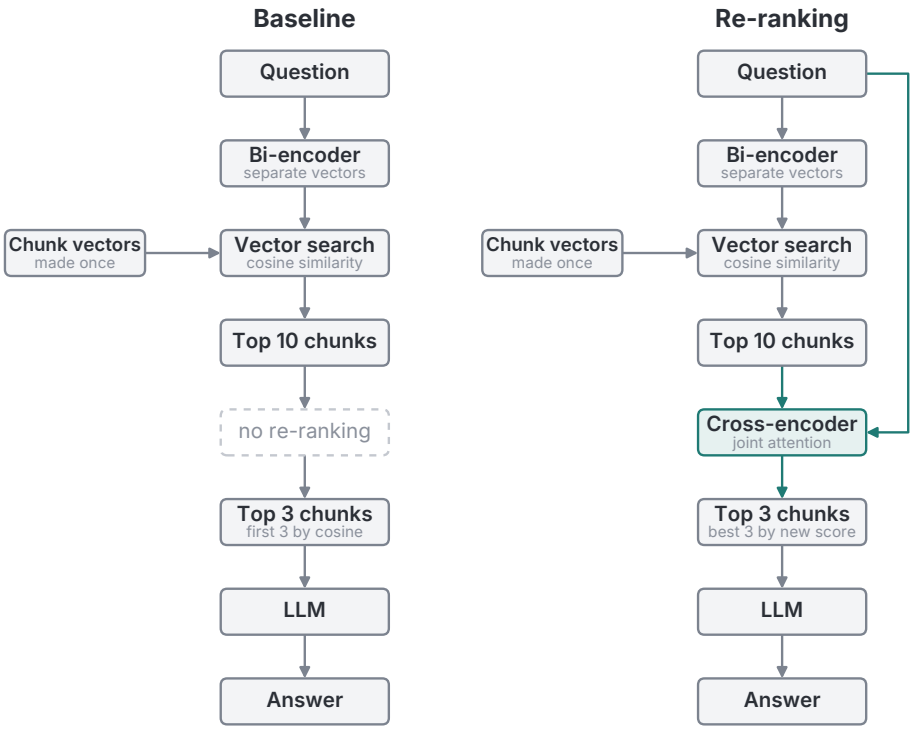
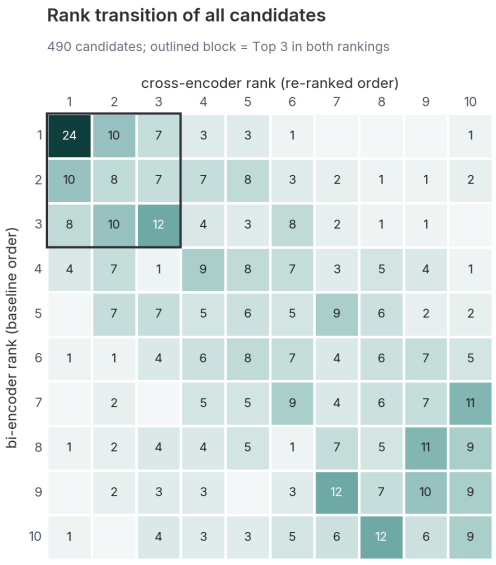


Figure 1. The two pipelines side by side. Both start with the same bi-encoder search. The re-ranking pipeline adds the cross-encoder, which reads the question together with each of the 10 chunks. The cross-encoder path is in green.



(a) Bi-encoder score vs. cross-encoder score.



(b) Rank before (rows) and after (columns).

Figure 2. The two models on the same 490 candidates (49 questions, 10 each). Gold chunks are in teal.

ter re-ranking, paragraph 1 came first, and the re-ranked answer listed the real checks.

In Q27 the re-ranked context had Article 5(1)(g). So the re-ranked answer could say that labelling lawful datasets is still allowed. The baseline answer only had a recital. It made up a looser exception about hair colour. In Q02 and Q47 both answers were fine, because a recital said almost the same thing. But only the re-ranked answer had the real article.

Where it made things worse. In Q20 ("Do people have to be told they are chatting with a bot?") the right clause fell from rank 1 to rank 5. The question and the law use different words, and the cross-encoder preferred three recitals. In Q46 (finding a missing child) it fell from rank 2 to rank 9. In Q23 the cross-encoder put the importer's 10 year rule first. The re-ranked answer then said providers must keep documents for 10 years "as per the obligations of importers". The number is right, but the source is for the wrong role. The last sentence of that answer was made up.

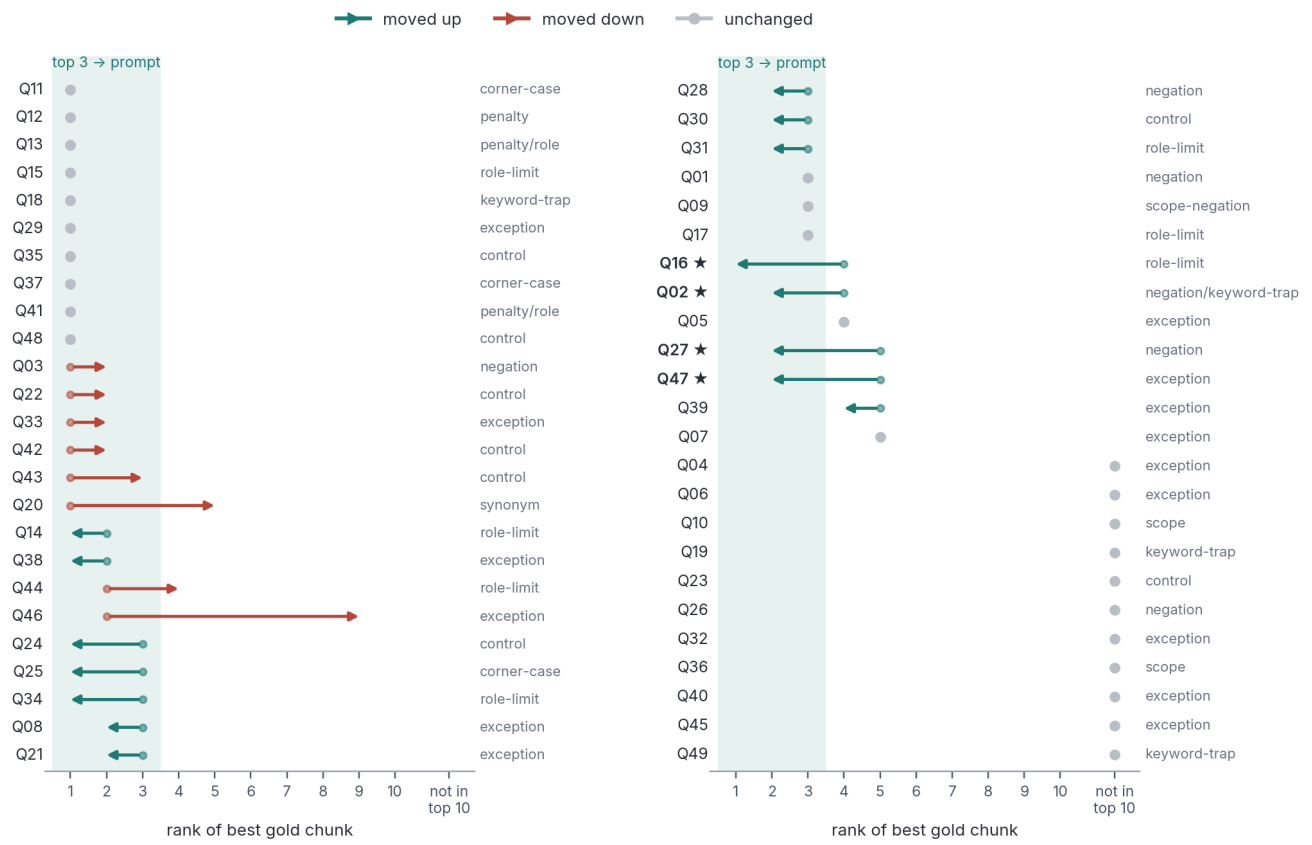


Figure 3. Rank of the right clause for every question. The arrow goes from the bi-encoder rank to the re-ranked rank. The green band is the top 3, which is what the LLM sees. A star marks a clause that moved from rank 4 or lower into the top 3.

When the first stage misses. In 11 of the 49 questions the right clause was not in the top 10 at all. The cross-encoder never saw it, so it could not help. The worst case is Q11, the fine for a start-up. The top 10 had Article 99(3), which says "whichever is higher". The start-up rule, Article 99(6), says "whichever is lower", but it was missing. So the re-ranked answer gave the wrong rule.

Recitals were a problem for both models. The top chunk was a recital for 25 questions before re-ranking and for 27 after. Recitals explain things in simple words, so they look like good answers. But they often leave out the exact condition or number. Figure 4 shows this, and also shows the results for each type of question.

Re-ranking helped most for role limits, where MRR went from 0.46 to 0.73. It also helped for negation, where hit@3 went from 0.6 to 0.8. Exceptions stayed the hardest type for both models. The reason is that the exception clause was often not in the top 10 at all. Recitals make up 45% of the chunks in the prompt before re-ranking and 43% after. So re-ranking did not remove them.

3 Latency Profiling

I timed each stage on an RTX 4060 laptop GPU. I did 5 warm-up runs, and then timed every question 3 times, which gives 147 runs. Table 3 has the results.

Re-ranking adds about 17 ms on the GPU. So retrieval goes from 4.5 ms to 21.5 ms, which is about 5 times slower. On the CPU it adds about 121 ms, which is about 28 times slower. Almost all of this time is the cross-encoder running 10 full passes. But the LLM takes about 5 seconds to answer. So the extra time is less than 1% of what the user waits. Figure 5 shows every single run.

Stage (ms)	median	p95
Embed the question	4.35	4.65
FAISS search	0.11	0.14
Cross-encoder, 10 pairs, GPU	17.01	17.73
Cross-encoder, 10 pairs, CPU	120.91	140.18
Baseline retrieval	**4.47**	**4.75**
Re-ranking retrieval, GPU	**21.48**	**22.32**
Re-ranking retrieval, CPU	**125.38**	**144.77**
LLM answer (both)	5,416	9,460

Table 3. Time per question for each stage.

4 Scaling Question

It is too slow, and none of its work can be saved. For one question I ran it on all 747 chunks. It took 870 ms, which is 1.16 ms per chunk. The whole bi-encoder search took 4.6 ms. So it was about 190 times slower, and this is only one document. Figure 6 shows what happens when the collection grows. With 1 million chunks the cross-encoder would need about 20 minutes for one question. The bi-encoder needs about 0.2 seconds.

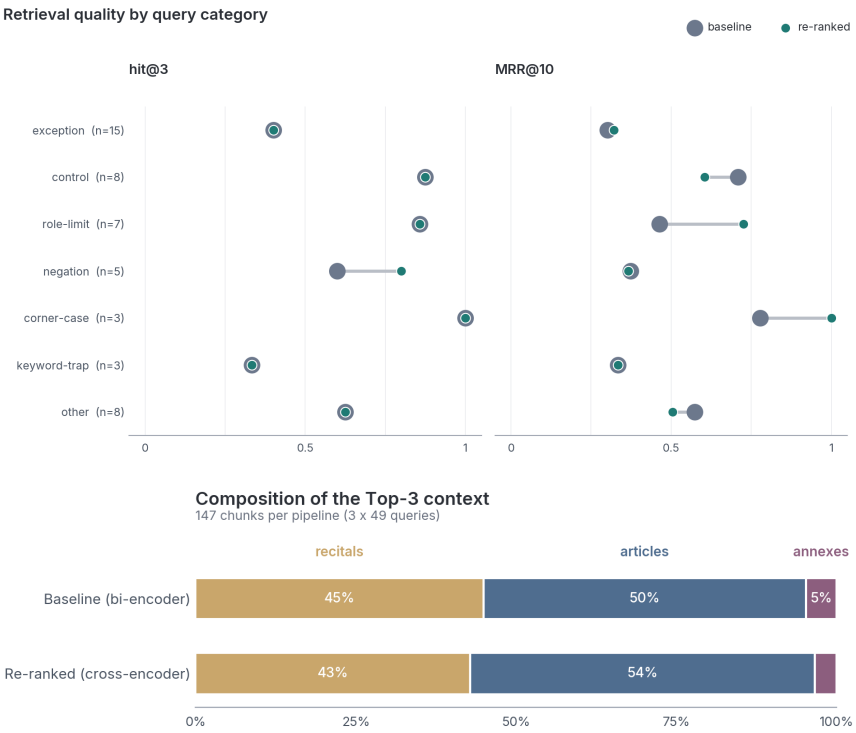
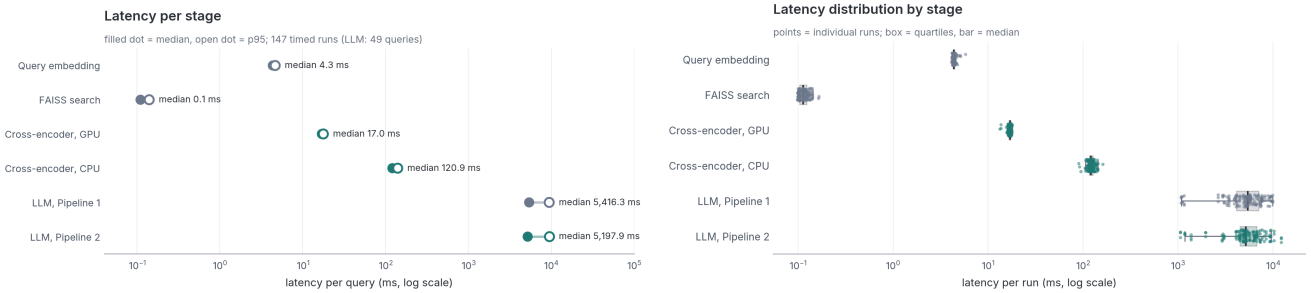


Figure 4. Top: hit@3 and MRR@10 for each type of question, before and after re-ranking. Bottom: how many of the chunks that reach the prompt are recitals, articles and annexes.



(a) Median (filled) and p95 (open).

(b) Every timed run.

Figure 5. Time per question for each stage, on a log scale. In the charts, Pipeline 1 is the baseline and Pipeline 2 is re-ranking.

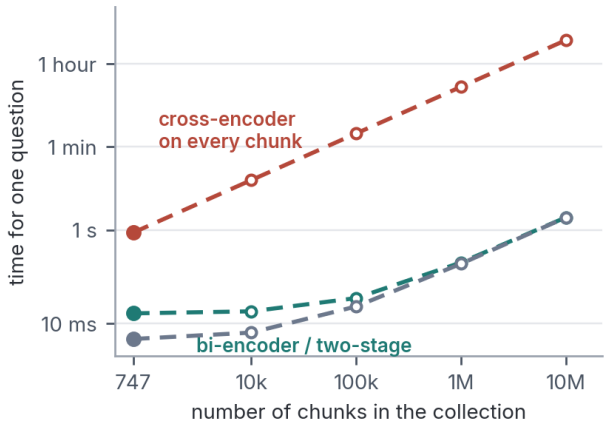


Figure 6. Time for one question as the collection grows. The first point is measured. The others are estimated by scaling up linearly.

The reason is simple. For the bi-encoder, all the chunk vectors are made once, before any question comes. For the cross-encoder, the score depends on the question and the chunk together. So nothing can be made early. Each new question needs one new pass for every chunk. The Sentence-BERT paper has a famous example. Finding the best pair among 10,000 sentences takes about 65 hours with BERT and about 5 seconds with SBERT.

So the normal way is to use two steps. A fast model picks a few candidates, and the slow model orders only those. This has one weakness, and I saw it. When I let the cross-encoder score all the chunks, it moved some missed clauses to rank 1. Q23 moved up from rank 41, Q06 from 16, and Q26 from 14. So the second step can only be as good as the first.

5 Ablation Study and Improved Retrieval Pipeline

Why. Section 2 showed two limits. The cross-encoder can only reorder what the bi-encoder found, and in 11 of 49 questions the right clause never reached it. And recitals

Change	MRR	Recall	Dev	Test
Reference (re-ranking)	0.502	0.776	0.506	0.498
Chunks of at most 120 tokens	0.558	0.796	0.577	0.539
Chunks of at most 400 tokens	0.444	0.653	0.503	0.387
No chunk header	0.496	0.735	0.505	0.487
Recitals removed	0.736	0.837	0.748	0.725
Re-rank the top 50	0.552	0.898	0.581	0.525
Add BM25 to the first stage	0.563	0.857	0.585	0.543
Bigger bi-encoder (bge-base)	0.560	0.796	0.569	0.551
Bigger re-ranker (MiniLM-L12)	0.537	0.776	0.545	0.530
All best changes (final pipeline)	**0.818**	**0.980**	**0.812**	**0.825**

Table 4. What I changed, one thing at a time. MRR = MRR@10 after re-ranking. Recall = share of questions where the right clause reached the cross-encoder. Dev and Test = MRR on Q01–Q24 and on Q25–Q49.

look like good answers to both models. So the text I index and the first stage may matter more than the re-ranker. I tested this by changing one design choice at a time.

How. Each run changes one choice and keeps the rest as in the re-ranking setup. I measured retrieval only, without the LLM, so each run takes a few seconds. I chose settings using Q01 to Q24 (dev) and used Q25 to Q49 (test) only to report. That way the final numbers are not tuned on the questions they are reported on. In Table 4, MRR is MRR@10 after re-ranking, and Recall is the share of questions where the right clause was among the candidates the cross-encoder saw.

Chunking. Chunks of at most 120 tokens raised MRR from 0.502 to 0.558. A small chunk covers one idea, so its vector matches a question better. Chunks of up to 400 tokens dropped it to 0.444. The bi-encoder reads only 256 tokens, and 216 of the 467 chunks were cut. The cut part could not be matched at all. The header also matters. Without the `[Article 99 – Penalties]` line, first-stage hit@3 fell from 0.633 to 0.469, because the header gives the article number and title to both encoders.

Which text to index. Removing the recitals gave the biggest single gain, with MRR at 0.736. Recitals say the same things as the articles in plain words, so they take up places in the top 10 that the binding article should have. One caution: my labels point to the law, so part of this gain is built in. That is why I also checked the answers, which improved too.

Candidates and first stage. Re-ranking the top 50 instead of the top 10 raised Recall from 0.776 to 0.898, because the right clause had more chances to reach the cross-encoder. BM25 is a keyword score. I merged its ranking with the bi-encoder ranking using reciprocal rank fusion: a chunk gets $1/(60 + \text{rank})$ from each list, and the two are added. This raised first-stage hit@3 from 0.633 to 0.694, because BM25 catches exact words like “start-ups” or “10 years”. A bigger bi-encoder (bge-base) or a bigger re-ranker (MiniLM-L12) helped only a little alone, with MRR at 0.560 and 0.537. They still worked on the same weak candidate list.

The final pipeline. I put the helpful changes together. The final pipeline:

1. indexes only the articles and annexes;
2. merges the bi-encoder and BM25 rankings and keeps 30 candidates;
3. re-ranks them with the MiniLM-L12 cross-encoder;
4. sends the best 3 to the same prompt.

Recall rose to 0.980, and MRR@10 went from 0.50 to 0.82. The test score of 0.825 is as good as the dev score of 0.812, so it is not fitted to the dev questions. The key facts in the answers went from 0.673 to 0.769 to 0.810 (Table 1). A larger re-ranker (bge-reranker-v2-m3) reached 0.828, but it did not fit on the 8 GB GPU next to the LLM. The final pipeline still fails on Q11. The cross-encoder put Article 99(6) at rank 8, just outside the top 3.

6 Summary

Re-ranking fixed real mistakes, mostly role limits and exceptions inside lists. It also broke a few cases, and it can never find a clause that the first stage missed. On this document the bigger problems were the recitals and the first stage, not the choice of re-ranker. The cross-encoder is cheap for 10 pairs, but it cannot be used on a whole collection.

Key points

- Re-ranking fixed four buried clauses and broke three cases.
- In 11 of 49 questions the first stage missed the right clause, and nothing could fix that later.
- Removing recitals and adding BM25 helped more than a bigger model did.

Limits. The key fact check is rough. For example, Q02 scores zero in every setup, because no answer uses the exact words “untargeted scraping”, even though all the answers are right. I used only 49 questions and one LLM, and the LLM does not give exactly the same answer every time.

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List of questions

The rank of the best right chunk for each question. "–" means it was not in the top 10. Base is the baseline (bi-encoder only), Re-rk is re-ranking the top 10, and Final is the final pipeline.

ID	Question	Gold	Base	Re-rk	Final
Q01	Is it allowed to use AI to detect the emotions of employees at their workplace?	Art 5(1)	3	3	1
Q02	Is it legal to build a face recognition database by scraping photos from the internet?	Art 5(1)	4	2	1
Q03	Can an AI system be used to predict whether a person will commit a crime?	Art 5(1)	1	2	1
Q04	When are police allowed to use real-time remote biometric identification in public spaces?	Art 5(1)	–	–	8
Q05	Can real-time biometric identification be used before a judge or authority has approved it?	Art 5(3)	4	4	2
Q06	Is every AI system listed in Annex III automatically considered high-risk?	Art 6(3)	–	–	1
Q07	Do providers of open-source general-purpose AI models have to prepare technical documentation?	Art 53(2)	5	5	9
Q08	Do deepfakes have to be labelled even when they are part of a satirical or artistic work?	Art 50(4)	3	2	1
Q09	Does the AI Act apply to AI systems used only for military purposes?	Art 2(3)	3	3	1
Q10	Does a person using an AI system for a personal hobby have obligations under the AI Act?	Art 2(10)	–	–	4
Q11	What is the maximum fine for a start-up that puts a prohibited AI system on the market?	Art 99(3) + Art 99(6)	1	1	1
Q12	What is the fine for giving misleading information to the authorities?	Art 99(5)	1	1	1
Q13	What fines can be imposed on providers of general-purpose AI models?	Art 101(1)	1	1	1
Q14	What are the obligations of a deployer, as opposed to a provider, of a high-risk AI system?	Art 26	2	1	1
Q15	When is a distributor or deployer treated as the provider of a high-risk AI system?	Art 25(1)	1	1	1
Q16	What must an importer check before placing a high-risk AI system on the market?	Art 23(1)	4	1	1
Q17	Which deployers must carry out a fundamental rights impact assessment?	Art 27(1)	3	3	2
Q18	How much training compute makes a general-purpose AI model count as having systemic risk?	Art 51(2)	1	1	1
Q19	From what date do the bans on unacceptable AI practices start to apply?	Art 113	–	–	–
Q20	Do people have to be told that they are chatting with a bot?	Art 50(1)	1	5	1
Q21	Is an AI system used for credit scoring considered high-risk?	Annex III pt 5	3	2	1
Q22	Does a person affected by a decision made with a high-risk AI system have a right to an explanation?	Art 86(1)	1	2	1
Q23	For how long must providers keep the technical documentation of a high-risk AI system?	Art 18(1)	–	–	1
Q24	What is the definition of an AI system?	Art 3(1)	3	1	1
Q25	Are SMEs and start-ups given lower fines under the AI Act?	Art 99(6)	3	1	1
Q26	Is it prohibited to use AI that exploits the vulnerabilities of children or elderly people?	Art 5(1)	–	–	1
Q27	Can a company use biometric data to sort people by their race or sexual orientation?	Art 5(1)	5	2	1
Q28	Is social scoring of citizens banned?	Art 5(1)	3	2	1
Q29	How quickly must a provider report a serious incident that caused a person's death?	Art 73(4)	1	1	1
Q30	What is the deadline for reporting a serious incident?	Art 73(2)	3	2	1
Q31	Does an employer have to inform workers before using a high-risk AI system in the workplace?	Art 26(7)	3	2	1
Q32	Can a high-risk AI system be used without completing the conformity assessment?	Art 46(1)	–	–	6
Q33	Do AI-generated news articles need to be disclosed if a human editor reviewed them?	Art 50(4)	1	2	1
Q34	Must people be informed when an emotion recognition system is used on them?	Art 50(3)	3	1	1
Q35	Is an AI tool that screens job applications and filters CVs high-risk?	Annex III pt 4	1	1	1
Q36	Does the AI Act apply to AI being developed and tested before it is released?	Art 2(8)	–	–	2
Q37	What happens to high-risk AI systems already on the market before the rules apply?	Art 111(2)	1	1	1
Q38	Does the deepfake disclosure obligation apply when the content is used by law enforcement to investigate crimes?	Art 50(4)	2	1	1
Q39	Are there exceptions to the ban on emotion recognition at work?	Art 5(1)	5	4	1
Q40	Can a provider avoid the high-risk obligations by documenting that its Annex III system is not risky?	Art 6(3) + Art 6(4)	–	–	1
Q41	What fine does a deployer face for not complying with its obligations?	Art 99(4)	1	1	1
Q42	Is AI used to assess students' exams or admission considered high-risk?	Annex III pt 3	1	2	1
Q43	Can AI be used to decide on asylum applications?	Annex III pt 7	1	3	1
Q44	Which uses of AI by law enforcement for lie detection are high-risk?	Annex III pt 6	2	4	2
Q45	Is facial recognition allowed for unlocking a phone or verifying identity?	Annex III pt 1	–	–	4
Q46	Can police use biometric identification to find a missing child?	Art 5(1)	2	9	5
Q47	Are AI systems used for detecting financial fraud classified as high-risk?	Annex III pt 5	5	2	1
Q48	How long do providers have to keep logs automatically generated by a high-risk AI system?	Art 19(1)	1	1	1
Q49	When will most of the AI Act's rules start to apply?	Art 113	–	–	–